

Secure Federated Maximum Entropy Classifier for Intrusion Detection in IoT Consumer Electronics

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Abstract—The rapid proliferation of connected consumer electronics within the Internet of Things (IoT) ecosystem has enabled a wide range of smart applications, but it has also exposed devices and networks to significant cybersecurity threats. Conventional intrusion detection systems based on centralized data collection suffer from critical limitations in terms of privacy preservation, scalability, and latency when deployed in large-scale IoT environments. To address these challenges, this paper proposes a Secure Federated Maximum Entropy Classifier (SFMEC) for IoT intrusion detection. The proposed SFMEC framework leverages federated learning to collaboratively train intrusion detection models across IoT gateways and edge servers, thereby enhancing scalability while preserving data privacy. Raw data generated by registered IoT devices is encrypted using agreed session keys and transmitted to their associated gateways, where local models are trained using a maximum entropy classifier. To improve security and robustness, both local and global models are protected using symmetric-key encryption and compressed using the ZIP algorithm to prevent parameter leakage and mitigate adversarial manipulation during model communication between IoT gateways and edge servers. Extensive experiments conducted on benchmark IoT intrusion detection datasets demonstrate that SFMEC consistently achieves higher detection accuracy and lower false-positive rates compared to existing approaches. The results highlight that combining entropy-based learning with secure federated optimization provides resilience against evolving threats and communication overheads; therefore, SFMEC enhances the security posture of next-generation IoT ecosystems.

Index Terms—IoT, federated learning, intrusion detection, maximum entropy classifier, security, performance analysis.

I. INTRODUCTION

ONE of the fast-growing and transformative field today is IoT, which seamlessly interconnects billions of

heterogeneous devices, including consumer electronics [1], [2]. By enabling real-time data acquisition, intelligent automation, and context-aware decision-making, IoT drives innovation across critical domains, including smart cities, precision agriculture, intelligent transportation, healthcare, smart homes, and advanced manufacturing [3], [4]. IoT consumer electronics are the Internet-connected devices that are designed to enhance convenience, efficiency, and control in everyday life. Some common examples include smart home devices (such as lights, thermostats, and security systems), wearable technology (like smartwatches and fitness trackers), and smart appliances. These IoT-enabled smart devices collect and exchange data, often leveraging edge computing, to support features such as remote control, automation, and personalized user experiences. Besides numerous useful applications, the IoT consumer electronics present substantial security and privacy risks due to weak authentication mechanisms, insecure data transmission, and outdated software. These devices frequently collect large volumes of sensitive information, which can be exposed to breaches, misuse, or unauthorized access if not adequately protected, leading to vulnerabilities in the network [5]. These issues become difficult to handle due to the highly distributed and resource-constrained nature of IoT infrastructure, and the presence of heterogeneous traffic [6], [7].

Traditional Intrusion Detection Systems (IDS), which rely heavily on centralized data collection and model training using machine learning algorithms, are not well-suited to IoT environments. Centralized approaches increase latency, incur high communication overhead, and expose raw data to privacy risks. Furthermore, IoT networks are characterized by heterogeneous devices and imbalanced traffic patterns, which degrade the effectiveness of conventional IDS solutions [8]. These limitations necessitate the development of decentralized, privacy-preserving, and adaptive intrusion detection frameworks tailored to IoT. To address these challenges, the emerging Federated Learning (FL) provides a promising paradigm by enabling collaborative model training across distributed IoT devices without sharing raw data [9]. However, standard FL frameworks remain vulnerable to inference attacks, model poisoning, and biases introduced by skewed data distributions [10]. Additionally, the focus in FL-based IDS remains on expensive deep learning approaches, such as FedAvg. However, other classifiers like the one based on the maximum entropy principle offer a mathematically grounded approach to robust classification under uncertainty, reducing overfitting as well as training time and improving generalization across diverse IoT data. When integrated with

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secure aggregation mechanisms, this combination enables the design of a Secure Federated Maximum Entropy Classifier (SFMEC), which ensures privacy, resilience, and scalability in IoT intrusion detection. However, such an approach has not been investigated in the literature yet, in spite of entropy-based identification of Distributed Denial of Service (DDoS) attacks in IoT [11] as well as entropy-based feature selection for intrusion detection [12].

A. Threat Model

We consider three adversarial threat models [13]: the Dolev–Yao (DY) threat model [14], the Canetti–Krawczyk (CK) threat model [15], and the Honest-but-Curious (HBC) model [16].

Under the DY threat model, the adversary has complete control over the public communication channels and can intercept, modify, delete, replay, and fabricate messages exchanged between network entities. The adversary can also impersonate any entity and initiate actions on its behalf. Moreover, entities may participate in multiple concurrent protocol executions initiated by the adversary.

In the CK threat model, the adversary retains all capabilities of the DY adversary and, in addition, can compromise session-specific secrets by accessing or hijacking session states during protocol execution.

In the HBC threat model, the adversary is assumed to be a legitimate participant who follows the protocol correctly but attempts to learn additional information by analyzing the messages exchanged during the protocol execution.

In addition, some IoT devices may be physically compromised because they are not continuously monitored (24×7), as is often the case in wireless sensor networks [17]. Consequently, an adversary can easily extract all credentials stored in the memory of these compromised devices.

B. Research Contributions

The major contributions of the paper include the following:

- A novel secure federated maximum entropy classifier is proposed, integrating entropy-based learning with federated optimization for IoT intrusion detection.
- The amount of data while exchanging models is reduced by applying appropriate compression mechanisms.
- The secure aggregation is performed to safeguard model updates against inference attacks and mitigate risks of model poisoning during communication.
- Extensive experiments on benchmark IoT intrusion detection datasets are conducted to demonstrate that SFMEC outperforms state-of-the-art federated IDS methods in terms of detection accuracy, false positive rate, and resilience to adversarial threats.

C. Paper Outline

The paper is organized into six sections. Some background literature that helps readers to follow the rest of the paper is contained in Section II. The related literature highlighting gaps and motivation for the proposed research work is summarized

in Section III. The new SFMEC framework is presented in Section IV. The two model aggregation algorithms (average and voting) are also outlined. The security analysis of the proposed scheme against various attacks is provided in Section V. The details and results of performance analysis are provided in Section VI. The last section (Section VII) contains the conclusion and future directions of this research work.

II. BACKGROUND

A. Federated Learning

FL is a distributed machine learning framework that enables multiple participants to collaboratively train a shared model while preserving data privacy and security [9]. In contrast to conventional centralized learning approaches, FL allows data owners to retain their data locally and perform training on their private datasets. Instead of sharing raw data, participants communicate only model updates, such as parameters or gradients, to a trusted aggregation server, which integrates these updates to construct a global model. The objective of the federated model is to achieve performance comparable to that of a centrally trained model that has access to the combined data from all participants. By leveraging locally trained updates from distributed data sources, federated learning aims to minimize the performance gap between the federated model and its centralized counterpart while ensuring data confidentiality. In FL terminology, data owners are commonly referred to as “nodes” or “clients”. FL is broadly categorized into two paradigms - horizontal FL, where clients share identical feature spaces but hold distinct data samples, and vertical FL, where clients possess disjoint feature sets across overlapping samples. The seminal algorithm FedAvg [9] pioneered the training of deep neural networks in a federated setting by simple averaging of weights sent by clients.

Building on FedAvg, subsequent research has sought to optimize communication efficiency, reduce indirect data leakage, and improve model convergence under non-IID conditions. Most existing studies emphasize deep neural networks and supervised learning. However, some advancements have begun broadening the scope of FL. For example, Gupta and Sharma [18] proposed FedCLUS, a single-shot federated clustering algorithm for distributed unlabeled data, and introduced FedCI, a cluster validity index specifically tailored for FL environments [19]. In the IoT domain, Sharma et al. [20] developed a federated decision tree learning algorithm capable of handling dynamic data streams, thereby enabling adaptive intelligence at the network edge.

At the same time, increasing attention has been devoted to secure federated learning, given the risks of inference attacks, model poisoning, and gradient leakage in adversarial environments. Approaches such as secure aggregation protocols [21], homomorphic encryption [22], and Differential Privacy (DP) [23] have been integrated into FL frameworks to safeguard model updates and preserve client confidentiality. More recently, robust aggregation mechanisms (e.g., Krum, Trimmed Mean, and Median-based defenses) have been investigated to mitigate the impact of poisoned updates from

compromised clients. While these techniques enhance privacy and resilience, they often incur significant computational and communication overhead, making them less practical for resource-constrained IoT devices. This gap highlights the need for lightweight, secure, and generalizable FL frameworks, motivating the exploration of entropy-based models in IoT intrusion detection.

B. Maximum Entropy (MaxEnt) Classifier

The MaxEnt classifier, based on information theory, is a probabilistic model that constructs the least biased distribution consistent with known constraints. According to Jaynes' maximum entropy principle, when multiple distributions satisfy the same constraints, the one with the highest entropy should be chosen to avoid unwarranted assumptions [24]. This property makes MaxEnt particularly effective in IDS, where malicious behaviors are diverse, dynamic, and often underrepresented in training data. In intrusion detection, MaxEnt models the conditional probability of a class label given an input feature vector, enabling flexible classification without assuming feature independence. Unlike rule-based or deterministic models, MaxEnt can adapt to high-dimensional, sparse, and imbalanced datasets, which are common in network security. Its probabilistic output also allows IDS to adjust decision thresholds, balancing between true positive rates and false alarms, a critical factor in large-scale deployments.

Let $\mathcal{D} = (x_n, y_n)_{n=1}^N$ with $x_n \in \mathcal{X}$, $y_n \in 1, \dots, M$ be the data points and class label respectively, and empirical input distribution $\hat{p}(x)$. Define feature functions $f_j(x, y)$, $j = 1, \dots, m$, and empirical constraints

$$\sum_x \hat{p}(x) \sum_y P(y | x) f_j(x, y) = \hat{E}[f_j], \quad j = 1, \dots, m, \quad (1)$$

where $\hat{E}[f_j] = \frac{1}{N} \sum_{n=1}^N f_j(x_n, y_n)$. The maximum entropy principle selects the conditional distribution $P(y | x)$ that maximizes the conditional entropy

$$H(P) = - \sum_x \hat{p}(x) \sum_y P(y | x) \log P(y | x) \quad (2)$$

subject to the moment-matching constraints (1) and per- x normalization $\sum_y P(y | x) = 1$ constraint. When optimization is carried out using Lagrange's multiplier method, the result turns out to be

$$P_{\theta}^*(y | x) = \frac{1}{Z_{\theta}^*(x)} \exp \left(\sum_{i=1}^m \theta_i f_i(x, y) \right), \quad (3)$$

where $\theta = (\theta_1, \dots, \theta_m)$ are Lagrange multipliers and

$$Z_{\theta}^*(x) = \sum_{y \in \mathcal{Y}} \exp \left(\sum_{i=1}^m \theta_i f_i(x, y) \right). \quad (4)$$

In the training phase, the parameters θ_i , $i = 1, \dots, m$, are estimated from the data.

Traditional deep learning IDS often demands significant computational power, making it impractical at the edge. MaxEnt, by contrast, is computationally efficient and capable of learning unbiased decision boundaries even from fragmented or incomplete IoT datasets. When integrated with federated

learning, MaxEnt enables collaborative training of intrusion detection models while preserving the privacy of device-level data, making it a promising option for building scalable and secure IoT IDS.

III. RELATED WORK

The related literature has been organized into two subsections to bring more clarity.

A. MaxEnt Classifier and IDS

The MaxEnt classifier has been widely explored in the design of IDS for computer networks due to its ability to model conditional class distributions under uncertainty without imposing strong independence assumptions. One of the earliest known works on the application of entropy in intrusion detection is by Lee and Xiang [25]. Afterwards, Gu et al. [26] proposed a new entropy-based metric, called intrusion detection capability, for measuring the performance of IDS. Later, Gu et al. [27] refined distribution-based anomaly detection using normalized entropy metrics, establishing its robustness against evolving traffic behaviors. Li et al. [28] applied a maximum entropy model to the KDD'99 dataset, showing that MaxEnt classifiers can achieve competitive classification performance in distinguishing normal and malicious traffic, particularly in cases of overlapping or imbalanced feature distributions. More recent studies integrated entropy analysis with machine learning classifiers for specific attack types, such as Distributed Denial-of-Service (DDoS) detection, where entropy features of packet flows significantly improved detection accuracy and reduced false alarms [29]. Entropy has also been leveraged as a classification head in hybrid intrusion detection frameworks. For instance, Yan et al. [30] utilized entropy within a transfer-learning architecture enabling adaptive network traffic classification across domains with shifting distributions.

In recent years, the problem of authenticated key agreement has received significant attention in the field of consumer electronics [31], [32], [33], [34], [35]. In the IoT security domain, entropy-driven approaches have also gained particular attention due to the heterogeneous and resource-constrained nature of IoT devices. Recently, A host-based light-weight IDS for IoT network has been proposed utilizing entropy and information related to packet transmission as features [36]. In another work, entropy-based uncertainty filtering is used to enhance the efficiency of classifiers in transfer learning based IoT IDS [37]. An IoT-IE approach combining information entropy of IoT traffic features is proposed to detect traffic anomalies in [38]. On a similar idea, a method to identify DDoS attacks in IoT is also proposed by analyzing the variation in entropy of traffic features [11]. This work also compares the performance of various generalized measures of entropy with Shannon's. In a recent work by Zhu et al. [39], a normalized Shannon entropy, quantifying the class imbalance at the client, is used to decide the contribution of a client in the aggregated global server in the FL server. Their proposed method is used to detect intrusions in IoT. In another work, Talpini et al. [40] utilized entropy of class labels to group IoT

devices for training a global model in an FL environment for IoT intrusion detection. However, the study of the federated entropy classifier remains unexplored in their scheme.

B. FL-Based IDS in IoT

With the rapid acceptance of FL because of its built-in advantages, efforts have been made to utilize it for designing effective intrusion detection methods in IoT. A notable early work is by Nguyen et al. who introduced DIoT, a federated self-learning anomaly detector at security gateways for learning device-specific normal behavior locally. Only model updates were shared with a global service server for aggregation. DIoT was evaluated on more than 30 devices (including Mirai infections) and reported 95.6% detection accuracy with 257 ms latency. Following this work, a federated deep learning scheme called DeepFed was proposed by Li et al. [41] for detecting threats in IoT networks using Convolutional Neural Networks (CNNs) and Gated Recurrent Units (GRUs). Zhang et al. [42] developed a FedIoT platform that implemented the FedDetect algorithm utilizing the Adam optimizer to detect anomalies across devices and validated its feasibility on a real IoT setup. After this, Rahman et al. [43] conducted a comparative study of centralized, distributed, and federated learning approaches for IoT intrusion detection. It was demonstrated that FL achieves performance close to centralized learning while outperforming traditional distributed methods. These studies underscore FL's viability as a scalable and privacy-conscious alternative for IDS in the IoT ecosystem.

Efforts also started to explore the efficiency of FL-driven IDS in specific application areas of IoT. Friha et al. proposed FELIDS, a targeted agricultural-IoT IDS, with three deep learning variants, viz. CNN, Recurrent Neural Network (RNN), and Deep Neural Network (DNN) for examining privacy-preserving gains on domain datasets [44]. The results showed better performance of FELIDS over centralized IDS. Belabri et al. emphasized pre-training to mitigate non-IID skew and class imbalance arising due to inherent heterogeneity of IoT traffic [45]. The results indicated that pre-trained global models significantly increase the F1-score over naive FedAvg. In a similar direction, another work worth mentioning is by Sun et al. who proposed FedMADE, a dynamic aggregation method that clusters clients by traffic patterns and weighs updates by contribution to global performance [46]. It was shown that FedMADE improved minority-attack detection by up to 71% with 4.7% extra per-round latency. A FL-based IDS utilizing knowledge distillation for improving computational efficiency was presented in [47]. In another work, Dey and Sharma proposed a change point detection-based method on activities of IoT devices to identify DDoS attacks [48].

Recently, Bouzinis et al. presented the StatAvg method, where clients share statistics like mean and variance of features of local data with the server, which, after aggregation, sends them back to all clients. The clients then use aggregated statistics for universal normalization [49]. It is to be highlighted that StatAvg can be applied before any FL method. A federated contrastive learning method, FeCO, has been proposed for more accurately identifying the benign class [50]. Lu et al. developed a personalized federated learning

approach using self-labeling for heterogeneous IoT intrusion detection, enabling each client to train personalized models despite diverse statistical distributions adaptively. Peng et al. [51] integrated FL with knowledge distillation for efficient training of a global model to detect IoT intrusion detection. Karunamurthy et al. [52] applied the Chimp optimization algorithm for selection of optimal features and CNN for training a global model in FL based framework of IoT intrusion detection.

Research Gap and Novelty: The decentralized nature of the IoT system makes it vulnerable to various attacks, such as model poisoning, where the malicious participants can manipulate their local models in order to compromise the global model's integrity. As a result, countermeasures, such as using a central server to identify and filter out anomalous participants based on their model performance, can help mitigate such kind of threats. A comparison of some prominent existing research is presented in Table I. Even though there is evidence of using entropy for IoT intrusion detection, its integration into federated and secure learning frameworks remains underexplored to the best of our knowledge, which leaves open opportunities for lightweight, privacy-preserving, and scalable IDS solutions. To mitigate these issues, we propose a secure federated maximum entropy classifier for intrusion detection in an IoT network.

IV. SFMEC: THE PROPOSED CLASSIFIER

A. Federated Maximum Entropy Classifier

The conditional maximum entropy model in (3) and (4) discussed in Section II-B can also be written in a compact form as

$$P_{\theta}(y | x) = \frac{\exp(\theta^T \mathbf{f}(x, y))}{Z_{\theta}(x)},$$

$$Z_{\theta}(x) = \sum_{y' \in \mathcal{Y}} \exp(\theta^T \mathbf{f}(x, y')). \quad (5)$$

The parameters θ can be fitted by minimizing the regularized negative log-likelihood

$$J(\theta) = \frac{1}{N} \sum_{i=1}^N (-\log P_{\theta}(y_i | x_i)) + \frac{\lambda}{2} \|\theta\|^2 \quad (6)$$

$$= -\frac{1}{N} \sum_{i=1}^N \theta^T \mathbf{f}(x_i, y_i) + \frac{1}{N} \sum_{i=1}^N \log Z_{\theta}(x_i) + \frac{\lambda}{2} \|\theta\|^2 \quad (7)$$

where the last term is for regularization to minimize overfitting of data, and λ is the regularization parameter. The gradient can be computed as

$$\nabla J(\theta) = -\frac{1}{N} \sum_{i=1}^N \mathbf{f}(x_i, y_i) + \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{p_{\theta}(y|x_i)}[\mathbf{f}(x_i, y)] + \lambda \theta. \quad (8)$$

The parameters θ can be computed by equating (8) to zero.

In an FL environment, the data is partitioned across K clients. Each client k has data $\mathcal{D}_k$ of size N_k such that

TABLE I
COMPARISON OF EXISTING SCHEMES ON INTRUSION DETECTION IN IIOT

Scheme	Approach Type	FL-based	Entropy-based	ML/DL-based	Secure Communication between IIOT Devices and Gateway
Lee and Xiang (2001) [25]	Entropy IDS	✗	✓	✗	✗
Gu et al. (2006) [26]	IDC Metric	✗	✓	✗	✗
Gu et al. (2009) [27]	Normalized Entropy AD	✗	✓	✗	✗
Li et al. (2010) [28]	MaxEnt Classifier	✗	✓	✓	✗
Dong et al. (2018) [29]	Hybrid Entropy-ML	✗	✓	✓	✗
li et al. (2019) [41]	FL with CNN+GRU	✓	✗	✓	✓
Yan et al. (2020) [30]	Entropy + Transfer Learning	✗	✓	✓	✗
Zhang et al. (2021) [42]	Adam optimizer	✓	✗	✓	✓
Rahman et al. (2021) [43]	FL vs CL vs DL Study	✓	✗	✓	✗
Katsura et al. (2021) [36]	Lightweight Entropy	✗	✓	✗	✗
Alturki et al. (2022) [37]	Entropy in TL	✗	✓	✓	✗
Sun et al. (2022) [38]	Traffic Entropy Features	✗	✓	✓	✗
Friha et al. (2022) [44]	FL with CNN/RNN/DNN	✓	✗	✓	✗
Belabri et al. (2022) [45]	Pre-trained FL	✓	✗	✓	✗
Pandey et al. (2023) [11]	Generalized Entropy	✗	✓	✓	✗
Sun et al. (2023) [46]	Dynamic Clustering FL	✓	✗	✓	✗
Benamer et al. (2024) [47]	Efficient FL	✓	✗	✓	✗
Bouzinis et al. (2025) [49]	Statistic Sharing FL	✓	✗	✓	✗
Wang et al. (2025) [50]	Federated Contrastive	✓	✗	✓	✗
Zhu et al. (2026) [39]	PFLSE Framework	✓	✓	✓	✗
Talpini et al. (2023) [40]	Clustered FL-IDS	✓	✓	✓	✗
Peng et al. (2025) [51]	FD-IDS Framework	✓	✗	✓	✗
Karunamurthy et al. (2025) [52]	Optimal FL-IDS	✓	✗	✓	✗
Proposed SFMEC	Integration of FL in MaxEnt classifier and security	✓	✓	✓	✓

$N = \sum_k N_k$. The global objective function then decomposes to

$$J(\theta) = \sum_{k=1}^K \frac{N_k}{N} J_k(\theta), \text{ and} \\ J_k(\theta) = \frac{1}{N_k} \sum_{(x,y) \in \mathcal{D}_k} (-\log P_\theta(y|x)) + \frac{\lambda}{2} \|\theta\|^2, \quad (9)$$

changing gradient (8) to

$$\nabla J(\theta) = \sum_{k=1}^K \frac{N_k}{N} \nabla J_k(\theta). \quad (10)$$

Therefore, the θ for the global model can be computed by taking a weighted average of the parameters from local models at the aggregation server.

B. Compression and Security Methods

The proposed SFMEC applies both compression/decompression methods and secure communications among the network entities.

1) *Compression and Decompression*: For compression of data, we use the ZIP algorithm that mainly uses the DEFLATE compression method. DEFLATE combines two techniques: 1) LZ77 and 2) Huffman coding. On the other hand, we use the UNZIP algorithm for decompression of the data. ZIP supports encryption of the data (for example, symmetric encryption using the Advanced Encryption Standard (AES-256) algorithm that produces 128-bit ciphertext on a 128-bit plaintext message using a 256-bit symmetric key), in order to secure the data inside the archive.

2) *Registration Phase*: The registration phase contains the registration of all the deployed IIOT devices, IIOT gateway node, and edge server node in the network. This is a one-time process that is performed by a Trusted Registration Authority (TRA) before deployment of the nodes in the IIOT network.

***IIOT Devices Registration**: The following steps are essential to register an IIOT device, say D_i , associated with an IIOT gateway node, GW_j :

Step 1: The TRA selects unique and district identity ID_{D_i} , a primary secret key MK_{D_i} of D_i , and a temporal identity TID_{D_i} .

Step 2: The TRA stores the registration information $\{ID_{D_i}, TID_{D_i}, MK_{D_i}\}$ into D_i 's memory, and also securely sends these credentials to the GW_j .

***IIOT Gateway and Edge Server Registration**: The following steps are needed to register an IIOT gateway node, GW_j , and an edge server, say ES_l :

Step 1: The TRA selects a unique and district identity ID_{GW_j} , and a random shared (symmetric) secret key SK_{GW_j, ES_l} between GW_j and ES_l .

Step 2: The TRA also selects a unique and district identity ID_{ES_l} .

Step 3: The TRA sends the registration information $\{ID_{GW_j}, SK_{GW_j, ES_l}\}$ securely to GW_j , while it also the registration credentials $\{ID_{ES_l}, SK_{GW_j, ES_l}\}$ securely to the ES_l .

3) *Secure Communication Between IIOT Devices and Gateway*: The following steps outline the procedure for establishing a session key SK_{D_i, GW_j} between an IIOT device and a gateway:

Step 1: The IIOT device D_i generates a random secret nonce rn_{D_i} and retrieves the current timestamp CT_{D_i} . It then computes $A_i = rn_{D_i} \oplus h(MK_{D_i} || CT_{D_i})$. Next, D_i transmits the message $\langle A_i, CT_{D_i} \rangle$ to the gateway GW_j .

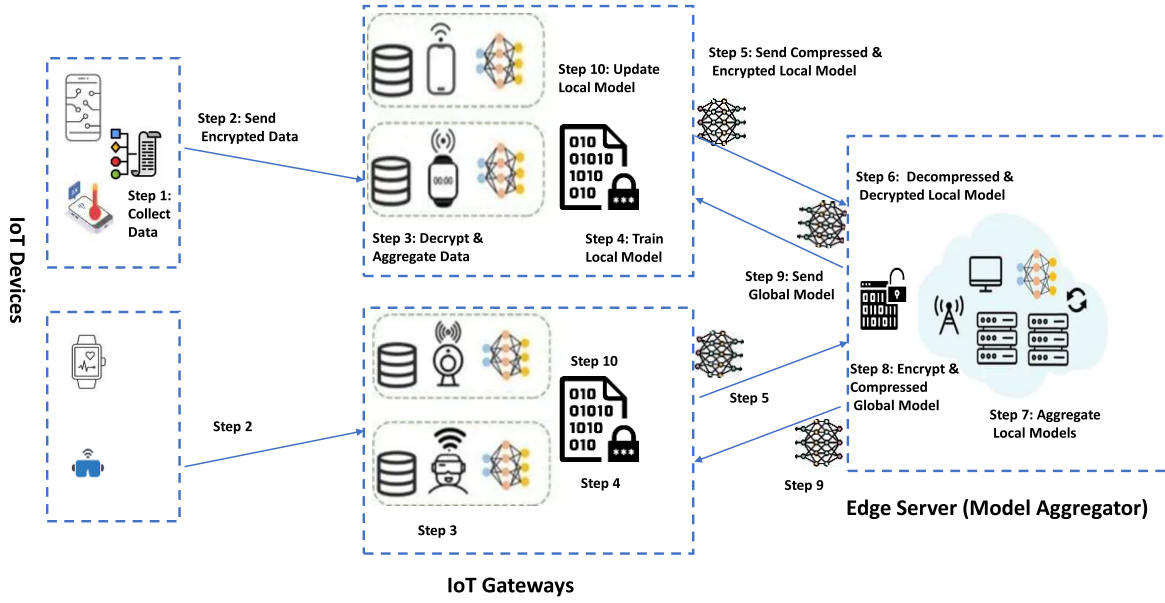


Fig. 1. Three-tier architecture with 10-step workflow for training SFMEC for intrusion detection in IoT.

Step 2: Upon receiving the message, GW_j first verifies its freshness by checking the timestamp condition: $|CT_{D_i}^* - CT_{D_i}| \leq \Delta T$, where ΔT represents the maximum permissible transmission delay and $CT_{D_i}^*$ denotes the reception time of the message. If the timestamp is valid, GW_j generates its own random secret nonce rn_{GW_j} and current timestamp CT_{GW_j} . It then computes the following:

$$\begin{aligned} rn_{D_i} &= A_i \oplus h(MK_{D_i} || CT_{D_i}), \\ B_i &= h(rn_{GW_j} || SK_{GW_j, ES_i}) \oplus h(MK_{D_i} || CT_{GW_j}), \\ SK_{GW_j, D_i} &= h(rn_{D_i} || h(rn_{GW_j} || SK_{GW_j, ES_i}) \\ &\quad || MK_{D_i} || CT_{D_i} || CT_{GW_j}), \end{aligned}$$

and its verifier $SKV = h(SK_{GW_j, D_i} || CT_{GW_j})$. Finally, GW_j sends the message $\langle B_i, SKV, CT_{GW_j} \rangle$ to D_i .

Step 3: After receiving $\langle B_i, SKV, CT_{GW_j} \rangle$, D_i verifies the timestamp CT_{GW_j} . If valid, it computes

$$\begin{aligned} h(rn_{GW_j} || SK_{GW_j, ES_i}) &= B_i \oplus h(MK_{D_i} || CT_{GW_j}), \\ SK_{D_i, GW_j} &= h(rn_{D_i} || h(rn_{GW_j} || SK_{GW_j, ES_i}) \\ &\quad || MK_{D_i} || CT_{D_i} || CT_{GW_j}), \\ SKV' &= h(SK_{D_i, GW_j} || CT_{GW_j}). \end{aligned}$$

If $SKV' = SKV$, both D_i and GW_j have successfully established the same session key $SK_{D_i, GW_j} = SK_{GW_j, D_i}$, which is then used for secure communication. Note that this established session key also connects the association with the edge server.

4) Secure Communication Between IoT Gateway and Edge Server: For secure communication between GW_j and ES_l , they will use the pre-shared symmetric key SK_{GW_j, ES_l} .

C. SFMEC for IoT Intrusion Detection

The proposed SFMEC comprises a federated maximum entropy classifier with secure compression and decompression as outlined in the previous two subsections. It can be applied

for intrusion detection in IoT by adopting a three-tier architecture consisting of the IoT devices layer, gateway layer, and edge server as illustrated in Fig. 1. For ensuring the scalability of the solution and considering the resource-constrained nature of consumer electronics, the data from devices is collected and stored at the IoT gateways securely. The gateways act as clients in the FL environment and train local models. The model training consists of a 10-step process along with a trusted edge server as an aggregator.

1) IoT Device Layer: The first tier comprises heterogeneous IoT consumer devices such as smartphones, wearable sensors, VR headsets, and embedded monitoring units.

- **Step 1:** The data is generated by these devices. These are tiny resource-constrained devices and cannot perform complex computation. The task of an IoT device D_i is to collect data and transmit it securely using the established session key SK_{D_i, GW_j} to the IoT gateway GW_j . Later, the GW_j decrypts the received data using the established session key SK_{GW_j, D_i} and then sends the encrypted data using the pre-shared symmetric key SK_{GW_j, ES_l} to its associated edge server ES_l .

2) IoT Gateways Layer: The second tier serves as an intermediate processing unit between the IoT devices and the edge server.

- **Step 2:** The data from multiple IoT devices within a specific geographic or logical domain is securely collected and aggregated at the gateway.
- **Step 3:** A local MaxEnt classifier is trained on the aggregated dataset, enabling near-real-time analytics without transferring raw data to the cloud.
- **Step 4:** The trained model parameters are compressed to reduce communication overhead and encrypted using the pre-shared symmetric key with the global server to protect confidentiality during transmission.

- **Step 5:** Finally, the encrypted model updates are transmitted securely to the edge server.

3) *Edge Server:* The third tier consists of a trusted edge server. This server manages and updates the global model.

- **Step 6:** The edge server decrypts the received model updates from all participating IoT gateways.
- **Step 7:** The federated MaxEnt aggregation algorithm is employed to integrate these updates into a unified global model, ensuring improved generalization across diverse data distributions.
- **Step 8:** The updated global model is again secured. The model is encrypted using the pre-shared symmetric key shared with the gateway ES_i .
- **Step 9:** The secured model is disseminated back to the IoT gateways for subsequent training rounds, completing the collaborative learning cycle.
- **Step 10:** Edge server transmits the updated global model to its corresponding IoT gateways, where the gateway replaces its current model with the newly received updated version to maintain synchronization with the global learning framework.

Steps 4 to 10 are repeated till model convergence is achieved,

The three-tier architecture effectively combines the advantages of edge computing and FL. Besides the averaging method of SFMEC, we have also implemented a voting-based aggregation of θ at the server for better comparison. It is pertinent to mention that voting is a simple and popular technique of combining models in ensemble learning. The step-wise process is also outlined in Algorithm 1.

V. SECURITY ANALYSIS

Based on the threat model defined in Section I-A, we show that the proposed scheme can resist the following attacks in an IoT environment.

1) Replay Attack

A replay attack is an attack in which an adversary captures a legitimate data transmission during the communications and later retransmits it to deceive a system into authorizing an unauthorized action.

The messages exchanged during the secure communication between an IoT device D_i and gateway GW_j include $\langle A_i, CT_{D_i} \rangle$ and $\langle B_i, SKV, CT_{GW_j} \rangle$. Even if these messages are intercepted by an adversary, say $\mathcal{A}$, and resent by the adversary $\mathcal{A}$ to the intended receivers, both D_i and GW_j can easily detect them by simply validating the freshness of the timestamps (CT_{D_i} and CT_{GW_j}) attached in the messages. In this way, these messages are simply dumped by D_i and GW_j . As a result, the attached current timestamps safeguard against replay attacks in the proposed scheme.

2) Man-in-the-Middle Attack

A Man-in-the-Middle (MitM) attack is an active attack in which an adversary covertly intercepts and potentially alters communication between two communicating entities, which

Algorithm 1 Secure Federated Maximum Entropy Classifier for IoT Intrusion Detection

Require: Dataset D , label column L
Ensure: Global_Averaged, and Global_Voting models
[SFMEC- Client side at IoT gateways]

- 1: **Initialize:** Load dataset D
- 2: Preprocessing of the dataset
- 3: Train MaxEnt classifier on the data
- 4: Extract weights θ
- 5: Compress and secure weights using steps in Section IV-B4
- 6: Send weights to edge server **[SFMEC- Server side at edge server]**
- 7: **for** each client i **do**
- 8: Securely decompress the weights as explained in Section IV-B4
- 9: **end for**
- 10: **[Global Average Model]**
- 11: Calculate the average of weights of all clients in θ_{avg}
- 12: Compute and store metrics (Only for performance analysis)
- 13: Securely compress the weights as explained in Section IV-B4
- 14: Send θ_{avg} to all clients
- 15: **[Global Voting Model (only for performance analysis)]**
- 15: Count the number of positive, negative, and zero values across clients
- 16: Assign weight + 1, -1, or 0 based on the majority sign
- 17: **return** voting-based weights θ_{vote}
- 18: Compute and store metrics

cause them to believe they are communicating directly with each other.

Assume that an adversary $\mathcal{A}$ intercepts the messages $\langle A_i, CT_{D_i} \rangle$ and $\langle B_i, SKV, CT_{GW_j} \rangle$, and attempts to modify these messages on the fly during the communication. To modify the message $\langle A_i, CT_{D_i} \rangle$, the adversary $\mathcal{A}$ can generate a new timestamp, say CT'_{D_i} , and a random nonce rn'_{D_i} . However, without having the secret MK_{D_i} , $\mathcal{A}$ can not compute the modified legal component $A'_i = rn'_{D_i} \oplus h(MK_{D_i} || CT'_{D_i})$. Similarly, $\mathcal{A}$ will fail to modify other message $\langle B_i, SKV, CT_{GW_j} \rangle$ without the secrets MK_{D_i} as well as SK_{GW_j, ES_i} . Thus, the man-in-the-middle attacks are resisted in the proposed scheme.

3) Impersonation Attack

An impersonation attack is a form of active attack in which a malicious entity (attacker) pretends to be a trusted individual, organization, or service to deceive victims into disclosing sensitive information or granting unauthorized access to systems.

In order to generate a valid message $\langle A_i, CT_{D_i} \rangle$ on behalf of the IoT device D_i , an adversary $\mathcal{A}$ must possess the secret key MK_{D_i} . Similarly, to generate the message $\langle B_i, SKV, CT_{GW_j} \rangle$ on behalf of GW_j , $\mathcal{A}$ must possess the secret keys MK_{D_i} and SK_{GW_j, ES_i} . It is then clear that $\mathcal{A}$ can not create forged messages on behalf of either D_i or GW_j . Therefore, the proposed scheme is secure against IoT device impersonation attacks as well as gateway impersonation attacks.

TABLE II
 DETAILS OF DATASETS

S. No.	Dataset	Total Data Points	Number of Features	Number of Classes	Types of Attacks
1.	CIC	1,048,576	80	2	Benign, Bot
2.	IOT-IDS	625,784	86	9	Mirai-Ack Flooding, DoS-SYN Flooding, Scan Port OS, Mirai-Host Brute Force, Mirai-UDP Flooding, Mirai-HTTP Flooding, MITM ARP Spoofing, Scan Host Port, Normal
3.	SDN	1,188,333	80	16	Probe, Botnet, Brute-Force, DDoS-UDP, DoS-ICMP, DoS-UDP, Exploitation, Malware, Mirai, R2L-IMAP, Recon-Packet Sniffing, Spoofing, Recon-Ping Sweep, UR2, Web-Based, Normal
4.	RAJ-24	145,123	77	5	Man-in-the-Middle, Distributed Denial of Service, Ransomware, Buffer Overflow attacks, Normal
5.	WUSTL	1,194,464	49	5	Backdoor Traffic, Reconnaissance Traffic, DoS Traffic, Command Injection Traffic, Normal Traffic

- IOT-IDS focuses on IoT-specific threats and attack patterns.
- SDN dataset includes a wide range of attacks targeting SDN environments.
- *Source*: Publicly available IoT cyber security datasets.

4) Physical IoT Device Capture Attack

According to our threat model in Section I-A, if an IoT device D_i is physically captured, all the credentials stored in its memory can be extracted using the power analysis attacks [53]. In that case, an adversary will have the secret credentials $\{ID_{D_i}, TID_{D_i}, MK_{D_i}\}$ of the compromised D_i . However, it is noted that these credentials are distinct and unique for every IoT device that is generated during the registration process. This means that the adversary D_i can only compromise the secure communication between the compromised D_i and its gateway, but not the secure communications among the non-compromised IoT devices and gateway. Thus, the proposed scheme provides unconditional security against node capture attacks [17], [54].

5) ESL Attack Under CK-Adversary Model

The Ephemeral Secret Leakage (ESL) attack in the Canetti-Krawczyk (CK) adversary model represents a serious security threat in which an adversary $\mathcal{A}$ compromises temporary session secrets (ephemeral keys) and leverages them to infer long-term private keys or session keys. In the proposed scheme, the session key is constructed by both D_i and GW_j as $SK_{GW_j, D_i} = h(rn_{D_i} || h(rn_{GW_j} || SK_{GW_j, ES_i}) || MK_{D_i} || CT_{D_i} || CT_{GW_j})$, where the ephemeral secrets are rn_{D_i} and rn_{GW_j} , and the long-term secrets are SK_{GW_j, ES_i} and MK_{D_i} . Thus, even if the ephemeral secrets are compromised using the CK-adversary model, the adversary $\mathcal{A}$ can not compute the session key SK_{GW_j, D_i} without having the long-term secrets. Hence, the proposed scheme is secure against an ESL attack under the CK-adversary model.

VI. PERFORMANCE ANALYSIS

This section contains the results of extensive performance analysis of the proposed SFMEC for IoT intrusion detection. We have chosen five widely used datasets from the related literature. The description of the datasets is provided in Table II. Each dataset is processed following the steps outlined in Fig. 2. For benchmarking, we have also compared the SFMEC with a federated logistic regression classifier that was presented in [55] and with the popular FedAvg algorithm.

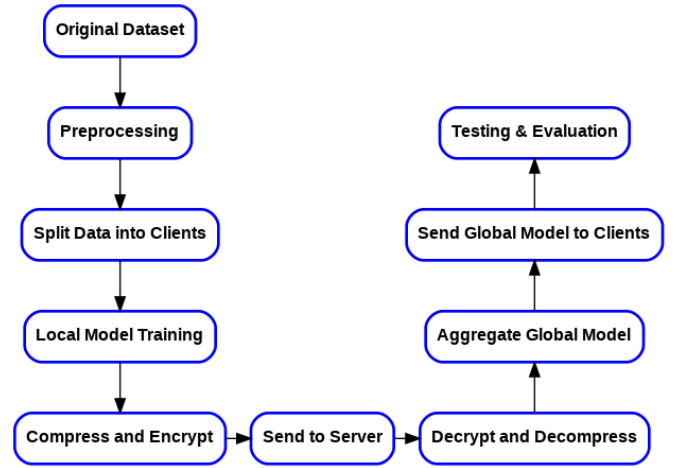


Fig. 2. Flow of data processing in experiments.

A. Metrics

The performance of SFMEC is assessed using five evaluation metrics for each dataset. In classification tasks, a confusion matrix is constructed, which serves as the basis for computing these metrics. Among them, accuracy is the most widely used. It represents the ratio of correctly predicted instances to the total number of predictions. Formally, accuracy is expressed as:

$$\text{Accuracy} = \frac{\text{True positives} + \text{True negatives}}{\text{Total number of predictions}}. \quad (11)$$

A high accuracy score reflects that the model correctly classifies a large proportion of instances. Accuracy is most appropriate when the dataset has a balanced class distribution and when the cost of different types of errors is comparable. However, in imbalanced datasets where one class significantly outweighs the others, accuracy alone can be misleading. In such scenarios, alternative metrics such as precision, recall, and F1-score offer a more reliable evaluation of model performance. The mathematical expression of the F1-score metric is

$$\text{F1-score} = \frac{2}{1/\text{Precision} + 1/\text{Recall}}, \quad (12)$$

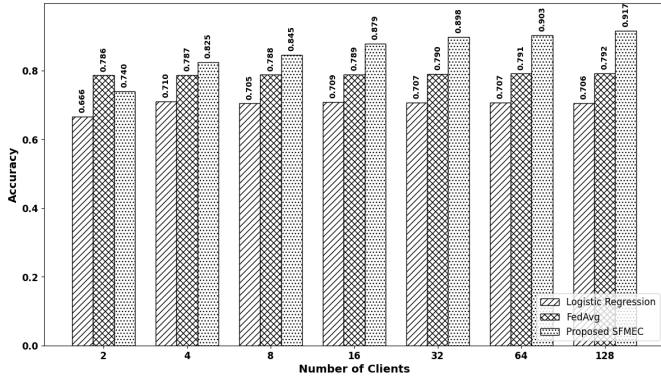


Fig. 3. Variation in accuracy with number of clients, i.e., IoT gateways for CIC dataset.

where

$$\text{Precision} = \frac{\text{True positives}}{\text{True positives} + \text{False positives}}, \quad (13)$$

and

$$\text{Recall} = \frac{\text{True positives}}{\text{True positives} + \text{False negatives}}. \quad (14)$$

The F1-score ranges between zero and one, where a value of one represents perfect precision (i.e., Type 1 errors) and recall (i.e., Type II errors). This measure is especially well-suited for applications in which misclassification costs are high or unevenly distributed across error types. Accordingly, it is frequently employed in problem domains such as fraud detection and intrusion detection.

To enable a more detailed assessment of classification performance, log-loss, commonly referred to as cross-entropy loss, is frequently utilized. In contrast to metrics such as accuracy, precision, recall, and the F1-score, which evaluate only the correctness of predicted class labels, log-loss incorporates the full class probability estimates produced by the model. This metric assigns higher penalties to incorrect predictions made with high confidence, while less certain errors incur smaller penalties, thereby offering a more informative characterization of predictive behavior. The log-loss metric is...

$$\text{log-loss} = -y \log p + (1 - y) \log(1 - p) \quad (15)$$

where y signifies the true binary label and p designates the probability of the predicted class. Lower values of log-loss correspond to improved predictive performance, with the optimal value of zero achieved when the estimated class probabilities align exactly with the ground-truth labels.

B. Results and Discussion

The experimental results comparing the proposed SFMEC with the federated logistic regression classifier are illustrated in Figs. 3–7, showing the variation of the accuracy metric (11) with respect to the number of clients (i.e., IoT gateways in the considered architecture described in Fig. 1). It can be noted that the accuracy of SFMEC remains consistently higher than that of the federated logistic regression classifier and is the same as the FedAvg classifier. This validates the efficacy of the proposed SFMEC. When compared with deep neural

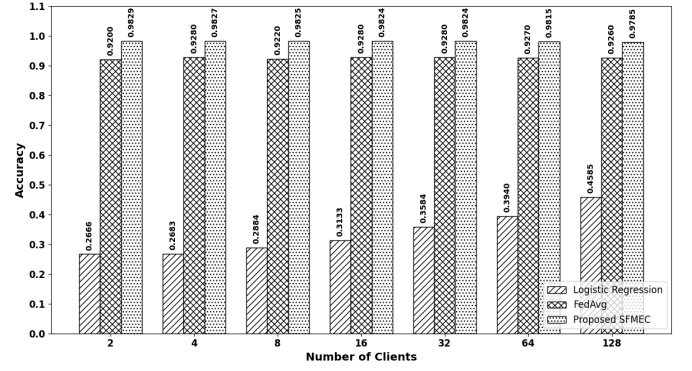


Fig. 4. Variation in accuracy with number of clients, i.e., IoT gateways for IOT-IDS dataset.

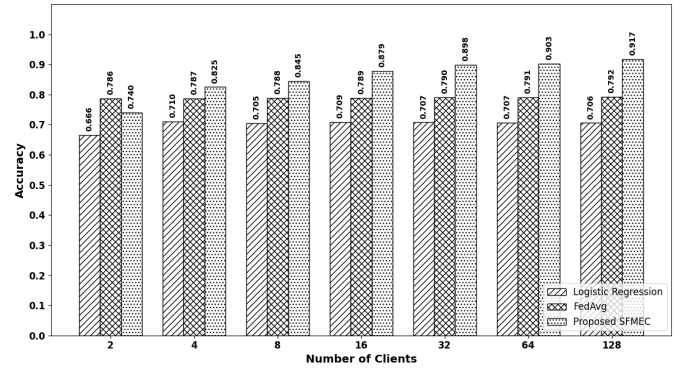


Fig. 5. Variation in accuracy with number of clients, i.e., IoT gateways for SDN dataset.

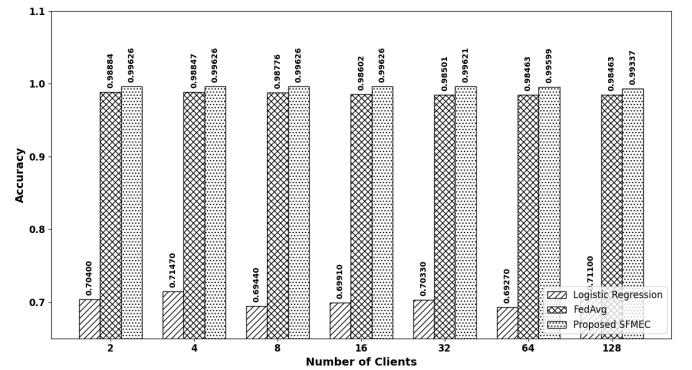


Fig. 6. Variation in accuracy with number of clients, i.e., IoT gateways for RAJ-24 dataset.

network-based FedAvg, the SFMEC offers advantages as it can provide better training over a small dataset and requires less training time.

With an intention to strengthen the performance of SFMEC, the values of accuracy, F1-score, and log-loss metrics for varying clients are shown in Tables III–VII comparing aggregation of SFMEC using averaging and voting methods. One limitation of voting-based aggregation, compared with averaging, is its tendency to yield higher log-loss values as the number of participating clients increases. Further, the performance results are strengthened statistically using the Analysis of Variance

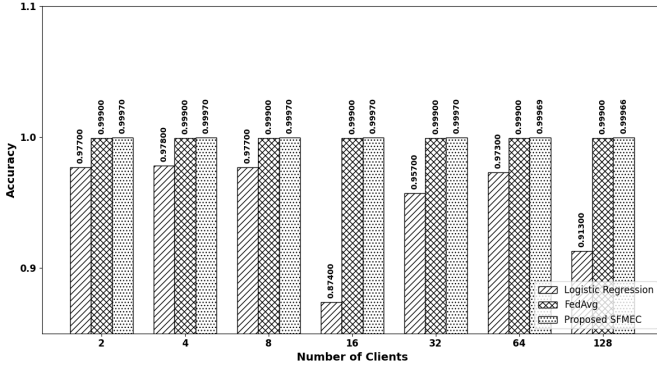


Fig. 7. Variation in accuracy with number of clients, i.e., IoT gateways for WUSTL dataset.

TABLE III
RESULTS OF SFMEC PERFORMANCE ON CIC DATASET

No. of Clients	Model Aggregator	Accuracy	F1-Score	Log-Loss
2	Averaging	0.82495	0.82247	0.35106
2	Voting	0.82147	0.82315	6.43493
4	Averaging	0.84531	0.84279	0.35004
4	Voting	0.83878	0.83989	5.81082
8	Averaging	0.87854	0.87706	0.35411
8	Voting	0.86327	0.86356	4.92841
16	Averaging	0.91656	0.91595	0.34981
16	Voting	0.90173	0.90174	3.54210
32	Averaging	0.73958	0.72838	0.52657
32	Voting	0.73751	0.72662	9.46119
64	Averaging	0.89817	0.89583	0.37710
64	Voting	0.90054	0.89879	3.58486
128	Averaging	0.90284	0.89984	0.36474
128	Voting	0.90284	0.89984	3.50187

TABLE IV
RESULTS OF SFMEC PERFORMANCE ON IoT-IDS DATASET

No. of Clients	Model Aggregator	Accuracy	F1-Score	Log-Loss
2	Averaging	0.98294	0.98206	0.06643
2	Voting	0.98434	0.98362	0.56443
4	Averaging	0.98274	0.98184	0.07032
4	Voting	0.98411	0.98337	0.57264
8	Averaging	0.98253	0.98161	0.07642
8	Voting	0.98325	0.98241	0.60388
16	Averaging	0.98241	0.98148	0.08383
16	Voting	0.98229	0.98135	0.63826
32	Averaging	0.98240	0.98147	0.09236
32	Voting	0.98159	0.98056	0.66365
64	Averaging	0.98151	0.98047	0.10321
64	Voting	0.97891	0.97754	0.76013
128	Averaging	0.97849	0.97705	0.11730
128	Voting	0.97490	0.97289	0.90466

(ANOVA) test by constructing the null hypothesis

$$H_0 : \mu_{\text{Algo1}} = \mu_{\text{Algo2}}$$

and the alternative hypothesis

$$H_1 : \mu_{\text{Algo1}} \neq \mu_{\text{Algo2}},$$

where Algo1 and Algo2 can be SFMEC, FedAvg, or LR (Logistic Regression). The results of this test are presented in Table VIII. For the majority of pairwise algorithm comparisons, the null hypothesis is rejected, indicating a statistically significant difference in mean performance at the 95% confidence level. In a limited number of cases, the performance of FedAvg and the proposed SFMEC method is found to be statistically the same. The values of lower and upper Confidence Interval (CI) are also included in Table VIII.

TABLE V
RESULTS OF SFMEC PERFORMANCE ON SDN DATASET

No. of Clients	Model Aggregator	Accuracy	F1-Score	Log-Loss
2	Averaging	0.98997	0.98995	0.05895
2	Voting	0.98993	0.98991	0.36299
4	Averaging	0.98981	0.98979	0.06686
4	Voting	0.98941	0.98939	0.38174
8	Averaging	0.98958	0.98956	0.07795
8	Voting	0.98864	0.98862	0.40957
16	Averaging	0.98935	0.98933	0.09047
16	Voting	0.98772	0.98770	0.44268
32	Averaging	0.98860	0.98858	0.10460
32	Voting	0.98696	0.98694	0.46993
64	Averaging	0.98739	0.98737	0.12053
64	Voting	0.98547	0.98546	0.52354
128	Averaging	0.98528	0.98526	0.13690
128	Voting	0.98452	0.98451	0.55782

TABLE VI
RESULTS OF SFMEC PERFORMANCE ON RAJ-24 DATASET

No. of Clients	Model Aggregator	Accuracy	F1-Score	Log-Loss
2	Averaging	0.99626	0.99610	0.02561
2	Voting	0.99626	0.99610	0.13464
4	Averaging	0.99626	0.99610	0.02868
4	Voting	0.99626	0.99610	0.13464
8	Averaging	0.99626	0.99610	0.03202
8	Voting	0.99626	0.99610	0.13464
16	Averaging	0.99626	0.99610	0.03552
16	Voting	0.99626	0.99610	0.13464
32	Averaging	0.99621	0.99603	0.03922
32	Voting	0.99621	0.99603	0.13665
64	Averaging	0.99599	0.99579	0.04313
64	Voting	0.99593	0.99573	0.14670
128	Averaging	0.99337	0.99279	0.04867
128	Voting	0.99442	0.99403	0.20096

TABLE VII
RESULTS OF SFMEC PERFORMANCE ON WUSTL DATASET

No. of Clients	Model Aggregator	Accuracy	F1-Score	Log-Loss
2	Averaging	0.99974	0.99974	0.00466
2	Voting	0.99973	0.99973	0.00980
4	Averaging	0.99971	0.99971	0.00498
4	Voting	0.99973	0.99973	0.00980
8	Averaging	0.99971	0.99971	0.00552
8	Voting	0.99970	0.99970	0.01066
16	Averaging	0.99971	0.99971	0.00745
16	Voting	0.99969	0.99969	0.01100
32	Averaging	0.99970	0.99970	0.01092
32	Voting	0.99969	0.99969	0.01117
64	Averaging	0.99969	0.99969	0.01396
64	Voting	0.99969	0.99969	0.01134
128	Averaging	0.99966	0.99966	0.01656
128	Voting	0.99967	0.99967	0.01203

Unlike traditional federated models that suffer degradation in accuracy due to data heterogeneity and communication noise, SFMEC maintains stability by optimizing the decision boundary through a maximum entropy principle. This leads to more balanced model updates from heterogeneous clients, thereby improving global model convergence. Additionally, the secure aggregation mechanism integrated within SFMEC ensures that sensitive device-level data remains confidential, preserving privacy without compromising model performance. The scalability of SFMEC is evident from its steady accuracy gains with an increasing number of clients. This suggests that the entropy-driven regularization promotes robustness against statistical diversity among IoT gateways. Furthermore, the

TABLE VIII
RESULTS OF STATISTICAL ANOVA TEST

Dataset	Metric	Algo1	Algo2	Mean Difference	p-value	Lower CI	Upper CI	Decision
CIC	Accuracy	FedAvg	LR	-0.0877	0.0008	-0.1378	-0.0376	Reject H_0
		FedAvg	SFMEC	0.0690	0.0066	0.0189	0.1191	Reject H_0
		LR	SFMEC	0.1567	< 0.001	0.1066	0.2068	Reject H_0
	F1-score	FedAvg	LR	-0.2131	< 0.001	-0.2675	-0.1586	Reject H_0
		FedAvg	SFMEC	-0.0471	0.0969	-0.1015	0.0073	Accept H_0
		LR	SFMEC	0.1660	< 0.001	0.1115	0.2204	Reject H_0
	Log-Loss	FedAvg	LR	0.4601	< 0.001	0.4062	0.5140	Reject H_0
		FedAvg	SFMEC	0.2619	< 0.001	0.2081	0.3158	Reject H_0
		LR	SFMEC	-0.1982	< 0.001	-0.2520	-0.1443	Reject H_0
IoT-IDS	Accuracy	FedAvg	LR	-0.5902	< 0.001	-0.6469	-0.5335	Reject H_0
		FedAvg	SFMEC	0.0563	0.052	-0.0004	0.1130	Accept H_0
		LR	SFMEC	0.6465	< 0.001	0.5898	0.7032	Reject H_0
	F1-score	FedAvg	LR	-0.5599	< 0.001	-0.6291	-0.4907	Reject H_0
		FedAvg	SFMEC	0.0131	0.8795	-0.0561	0.0824	Accept H_0
		LR	SFMEC	0.5731	< 0.001	0.5038	0.6423	Reject H_0
	Log-Loss	FedAvg	LR	4.5720	0.0002	2.2635	6.8804	Reject H_0
		FedAvg	SFMEC	0.0800	0.9957	-2.2285	2.3885	Accept H_0
		LR	SFMEC	-4.4920	0.0003	-6.8005	-2.1835	Reject H_0
SDN	Accuracy	FedAvg	LR	-0.3331	< 0.001	-0.4354	-0.2308	Reject H_0
		FedAvg	SFMEC	-0.0004	0.9999	-0.1027	0.1019	Accept H_0
		LR	SFMEC	0.3327	< 0.001	0.2304	0.4350	Reject H_0
	F1-score	FedAvg	LR	-0.3626	< 0.001	-0.4773	-0.2479	Reject H_0
		FedAvg	SFMEC	-0.0005	0.9999	-0.1152	0.1143	Accept H_0
		LR	SFMEC	0.3621	< 0.001	0.2474	0.4769	Reject H_0
	Log-Loss	FedAvg	LR	1.3678	0.0014	0.5451	2.1905	Reject H_0
		FedAvg	SFMEC	0.0456	0.9890	-0.7771	0.8683	Accept H_0
		LR	SFMEC	-1.3222	0.0018	-2.1449	-0.4995	Reject H_0
Raj-24	Accuracy	FedAvg	LR	-0.2837	< 0.001	-0.2904	-0.2771	Reject H_0
		FedAvg	SFMEC	0.0093	0.0056	0.0027	0.0159	Reject H_0
		LR	SFMEC	0.2931	< 0.001	0.2864	0.2997	Reject H_0
	F1-score	FedAvg	LR	-0.1804	< 0.001	-0.1852	-0.1757	Reject H_0
		FedAvg	SFMEC	0.0096	0.0002	0.0049	0.0144	Reject H_0
		LR	SFMEC	0.1901	< 0.001	0.1853	0.1949	Reject H_0
	Log-Loss	FedAvg	LR	0.6827	< 0.001	0.5902	0.7752	Reject H_0
		FedAvg	SFMEC	-0.0248	0.7759	-0.1173	0.0677	Accept H_0
		LR	SFMEC	-0.7075	< 0.001	-0.8000	-0.6150	Reject H_0
WUSTL	Accuracy	FedAvg	LR	-0.0488	0.0030	-0.0809	-0.0166	Reject H_0
		FedAvg	SFMEC	0.0007	0.9983	-0.0314	0.0328	Accept H_0
		LR	SFMEC	0.0495	0.0027	0.0173	0.0816	Reject H_0
	F1-score	FedAvg	LR	-0.0413	0.0012	-0.0659	-0.0167	Reject H_0
		FedAvg	SFMEC	0.0007	0.9971	-0.0239	0.0253	Accept H_0
		LR	SFMEC	0.0420	0.0011	0.0174	0.0666	Reject H_0
	Log-Loss	FedAvg	LR	0.4682	< 0.001	0.4642	0.4721	Reject H_0
		FedAvg	SFMEC	0.0080	0.0002	0.0041	0.0119	Reject H_0
		LR	SFMEC	-0.4602	< 0.001	-0.4641	-0.4562	Reject H_0

results indicate that SFMEC requires fewer communication rounds to achieve convergence compared to the baseline, highlighting its efficiency in resource-constrained IoT networks.

The findings thus validate the effectiveness of SFMEC as a privacy-preserving, scalable, and accurate intrusion detection framework for consumer IoT devices. By integrating security and entropy-aware learning within a federated setup, SFMEC bridges the gap between local model reliability and global network security, offering a promising direction for next-generation distributed intrusion detection systems in large-scale IoT ecosystems.

VII. CONCLUSION

The paper has presented a secure federated maximum entropy classifier for intrusion detection in an IoT consumer electronics network. The proposed framework combines federated learning with the maximum entropy principle to overcome limitations of centralized intrusion detection systems, particularly those related to data privacy, scalability, and class imbalance. By enabling IoT gateways to train local models securely and collaboratively without sharing raw data, SFMEC ensures privacy preservation while maintaining high detection performance, besides making IDS scalable. The entropy-based learning approach enhances robustness against data heterogeneity and uncertainty, leading to more balanced decision boundaries. In addition, the integration of secure aggregation and model compression mechanisms prevents information

leakage and reduces communication overhead, making the system practical for large-scale IoT environments. Experimental evaluations on benchmark IoT datasets of intrusion detection, along with comparison with popular FedAvg and federated logistic regression, demonstrate that SFMEC achieves the same or superior detection accuracy and lower false-positive rates even in identifying multiple types of attacks.

The SFMEC can establish a scalable, privacy-preserving, and resilient solution for next-generation IoT intrusion detection, offering a strong foundation for secure and trustworthy IoT network management. The performance of SFMEC in real IoT setups can be evaluated in the future. Further, the dynamic entropy tuning mechanisms, allowing the classifier to automatically adjust entropy regularization based on data heterogeneity and client reliability, can also be explored.

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REFERENCES

- [1] W. Lu et al., “Stones from other hills: Intrusion detection in statistical heterogeneous IoT by self-labeled personalized federated learning,” *IEEE Internet Things J.*, vol. 12, no. 10, pp. 14348–14361, May 2025.
- [2] S. Sharma, S. Kapoor, B. Srinivasan, and M. S. Narula, “HiCHO: Attributes based classification of ubiquitous devices,” in *Proc. Mobile Ubiquitous Syst., Comput.*, 2012, pp. 113–125.
- [3] P. Sun, S. Shen, Y. Wan, Z. Wu, Z. Fang, and X.-Z. Gao, “A survey of IoT privacy security: Architecture, technology, challenges, and trends,” *IEEE Internet Things J.*, vol. 11, no. 21, pp. 34567–34591, Nov. 2024.
- [4] S. Sharma, R. Kothari, and O. Ghaswi, “Data stream clustering using embedding dimension at edge gateway,” in *Proc. IEEE Int. Conf. Adv. Netw. Telecommun. Syst. (ANTS)*, Dec. 2022, pp. 213–218.
- [5] M. Shafiq, R. Yadav, A. R. Javed, and S. A. H. Mohsan, “CoopGBFS: A federated learning and game-theoretic-based approach for personalized security, recommendation in 5G beyond IoT environments for consumer electronics,” *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2648–2656, Feb. 2024.
- [6] S. Sharma and P. D. Bhatt, “Performance analysis of Gamma/M/1 model for IoT-based sensor data traffic,” *IEEE Wireless Commun. Lett.*, vol. 10, no. 11, pp. 2430–2434, Nov. 2021.
- [7] S. Sharma and P. Datt Bhatt, “Performance modeling of IoT-cloud gateway under diverse traffic characteristics,” *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 1, pp. 691–701, Feb. 2025.
- [8] B. B. Zarpel ao, R. S. Miani, C. T. Kawakani, and S. C. D. Alvarenga, “A survey of intrusion detection in Internet of Things,” *J. Netw. Comput. Appl.*, vol. 84, pp. 25–37, Apr. 2017.
- [9] J. Konečný, H. Brendan McMahan, F. X. Yu, P. Richtárik, A. Theertha Suresh, and D. Bacon, “Federated learning: Strategies for improving communication efficiency,” 2016, *arXiv:1610.05492*.
- [10] P. Kairouz et al., “Advances and open problems in federated learning,” *Foundations Trends Mach. Learn.*, vol. 14, nos. 1–2, pp. 1–210, 2020.
- [11] N. Pandey and P. K. Mishra, “Performance analysis of entropy variation-based detection of DDos attacks in IoT,” *Internet Things*, vol. 23, Oct. 2023, Art. no. 100812.
- [12] Z. Raeisi, H. R. Maleki, and R. Akbari, “An entropy-based multi-objective feature selection method for network intrusion detection,” *Cluster Comput.*, vol. 28, no. 12, p. 790, Nov. 2025.
- [13] A. Vangala, R. Maheshwari, A. K. Das, and S. Pal, “Cloud-assisted security framework for drone-enabled offshore communications,” in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, May 2023, pp. 1–6.
- [14] D. Dolev and A. Yao, “On the security of public key protocols,” *IEEE Trans. Inf. Theory*, vol. 29, no. 2, pp. 198–208, Mar. 1983.
- [15] R. Canetti and H. Krawczyk, “Universally composable notions of key exchange and secure channels,” in *Proc. Int. Conf. Theory Appl. Cryptograph. Techniques Adv. Cryptol. (EUROCRYPT)*, Amsterdam, The Netherlands, 2002, pp. 337–351.

- [16] A. Paverd, A. Martin, and I. Brown. (2014). *Modelling and Automatically Analysing Privacy Properties for Honest-but-Curious Adversaries*. Accessed: Jun. 2025. [Online]. Available: <https://ajpavard.org/publications/casper-privacy-report.pdf>
- [17] A. K. Das, "A random key establishment scheme for multi-phase deployment in large-scale distributed sensor networks," *Int. J. Inf. Secur.*, vol. 11, no. 3, pp. 189–211, Jun. 2012.
- [18] S. Gupta and S. Sharma, "Fedclus: Federated clustering from distributed homogeneous data," in *Proc. Int. Conf. Soft Comput. Eng. Appl.*, 2022, pp. 29–41.
- [19] S. Sharma and S. Gupta, "Evaluating the performance of FedCLUS algorithm using FedCI: A new federated cluster validity metric," *Social Netw. Comput. Sci.*, vol. 5, no. 4, p. 332, Mar. 2024.
- [20] S. Sharma, K. Arora, and P. S. P. Thakur, "Horizontal federating decision tree learning from data streams: Building intelligence in IoT edge networks," in *Proc. IEEE 8th World Forum Internet Things (WF-IoT)*, Oct. 2022, pp. 1–6.
- [21] K. Bonawitz et al., "Practical secure aggregation for privacy-preserving machine learning," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Oct. 2017, pp. 1175–1191.
- [22] L. T. Phong, Y. Aono, T. Hayashi, L. Wang, and S. Moriai, "Privacy-preserving deep learning via additively homomorphic encryption," *IEEE Trans. Inf. Forensics Security*, vol. 13, no. 5, pp. 1333–1345, May 2018.
- [23] R. C. Geyer, T. Klein, and M. Nabi, "Differentially private federated learning: A client level perspective," 2017, *arXiv:1712.07557*.
- [24] A. Golan and D. K. Foley, "Understanding the constraints in maximum entropy methods for modeling and inference," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3994–3998, Mar. 2023.
- [25] W. Lee and D. Xiang, "Information-theoretic measures for anomaly detection," in *Proc. IEEE Symp. Secur. Privacy. SP*, May 2000, pp. 130–143.
- [26] G. Gu, P. Fogla, D. Dagon, W. Lee, and B. Skorić, "Measuring intrusion detection capability: An information-theoretic approach," in *Proc. ACM Symp. Inf., Comput. Commun. Secur.*, Mar. 2006, pp. 90–101.
- [27] G. Gu, P. Porras, V. Yegneswaran, M. Fong, and W. Lee, "BotHunter: Detecting malware infection through IDS-driven dialog correlation," in *Proc. 16th USENIX Secur. Symp.*, 2007, p. 12.
- [28] Y. Li, B.-X. Fang, Y. Chen, and L. Guo, "A lightweight intrusion detection model based on feature selection and maximum entropy model," in *Proc. Int. Conf. Commun. Technol.*, Nov. 2006, pp. 1–4.
- [29] S. Dong and M. Sarem, "DDoS attack detection method based on improved KNN with the degree of DDoS attack in software-defined networks," *IEEE Access*, vol. 8, pp. 5039–5048, 2020.
- [30] F. Yan, G. Zhang, D. Zhang, X. Sun, B. Hou, and N. Yu, "TL-CNN-IDS: Transfer learning-based intrusion detection system using convolutional neural network," *J. Supercomput.*, vol. 79, no. 15, pp. 17562–17584, Oct. 2023.
- [31] T. Zhang, J. Shen, H. Yang, P. Vijayakumar, B. B. Gupta, and V. Arya, "Sustainable authentication and key agreement protocol using chaotic maps for industry 5.0," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 1580–1589, Feb. 2024, doi: [10.1109/TCE.2023.3339818](https://doi.org/10.1109/TCE.2023.3339818).
- [32] M. F. Ayub, X. Li, K. Mahmood, S. Shamshad, M. A. Saleem, and M. Omar, "Secure consumer-centric demand response management in resilient smart grid as industry 5.0 application with blockchain-based authentication," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 1370–1379, Feb. 2024, doi: [10.1109/TCE.2023.3320974](https://doi.org/10.1109/TCE.2023.3320974).
- [33] S. K. Behera, P. Kumar, D. P. Dogra, and P. P. Roy, "A robust biometric authentication system for handheld electronic devices by intelligently combining 3D finger motions and cerebral responses," *IEEE Trans. Consum. Electron.*, vol. 67, no. 1, pp. 58–67, Feb. 2021.
- [34] M. F. Ayub, S. Shamshad, K. Mahmood, S. H. Islam, R. M. Parizi, and K.-K.-R. Choo, "A provably secure two-factor authentication scheme for USB storage devices," *IEEE Trans. Consum. Electron.*, vol. 66, no. 4, pp. 396–405, Nov. 2020.
- [35] I. Ul Haq, J. Wang, and Y. Zhu, "An efficient authenticated key agreement scheme for consumer USB MSDs resilient to unauthorized file decryption," *IEEE Trans. Consum. Electron.*, vol. 65, no. 1, pp. 80–89, Feb. 2019.
- [36] Y. Katsura, A. Endo, I. Arai, and K. Fujikawa, "Efficient IDS for IoT networks using host-based data aggregation and multi-entropy analysis," *IEEE Access*, vol. 13, pp. 125406–125419, 2025.
- [37] B. Alturki and A. A. Alsulami, "Semi-supervised learning with entropy filtering for intrusion detection in asymmetrical IoT systems," *Symmetry*, vol. 17, no. 6, p. 973, Jun. 2025.
- [38] Y. Sun, J. Yu, J. Tian, Z. Chen, W. Wang, and S. Zhang, "IoT-IE: An information-entropy-based approach to traffic anomaly detection in Internet of Things," *Secur. Commun. Netw.*, vol. 2021, pp. 1–13, Dec. 2021.
- [39] X. Zhu et al., "PFLSE: A personalized federated learning framework based on Shannon entropy metric for intrusion detection in IIoT," *IEEE Internet Things J.*, vol. 13, no. 2, pp. 2936–2955, Jan. 2026.
- [40] J. Talpini, F. Sartori, and M. Savi, "A clustering strategy for enhanced FL-based intrusion detection in IoT networks," 2023, *arXiv:2307.14268*.
- [41] B. Li, Y. Wu, J. Song, R. Lu, T. Li, and L. Zhao, "DeepFed: Federated deep learning for intrusion detection in industrial cyber-physical systems," *IEEE Trans. Ind. Informat.*, vol. 17, no. 8, pp. 5615–5624, Aug. 2021.
- [42] Y. Zhang, B. Li, B. Liu, Y. Hu, and H. Zheng, "A privacy-aware PUFs-based multiserver authentication protocol in cloud-edge IoT systems using blockchain," *IEEE Internet Things J.*, vol. 8, no. 18, pp. 13958–13974, Sep. 2021.
- [43] S. A. Rahman, H. Tout, C. Talhi, and A. Mourad, "Internet of Things intrusion detection: Centralized, on-device, or federated learning?," *IEEE Netw.*, vol. 34, no. 6, pp. 310–317, Nov. 2020.
- [44] O. Friha, M. A. Ferrag, L. Shu, L. Maglaras, K.-K.-R. Choo, and M. Nafaa, "FELIDS: Federated learning-based intrusion detection system for agricultural Internet of Things," *J. Parallel Distrib. Comput.*, vol. 165, pp. 17–31, Jul. 2022.
- [45] O. Belarbi, T. Spyridopoulos, E. Anthi, I. Mavromatis, P. Carnelli, and A. Khan, "Federated deep learning for intrusion detection in IoT networks," 2023, *arXiv:2306.02715*.
- [46] S. Sun, P. Sharma, K. Nwodo, A. Stavrou, and H. Wang, "FedMADE: Robust federated learning for intrusion detection in IoT networks using a dynamic aggregation method," in *Proc. Int. Conf. Inf. Secur.*, 2024, pp. 286–306.
- [47] R. Benameur, A. Dahane, S. Souihi, and A. Mellouk, "A novel federated learning based intrusion detection system for IoT networks," in *Proc. ICC - IEEE Int. Conf. Commun.*, Jun. 2024, pp. 2402–2407.
- [48] M. Dey and S. Sharma, "Identification of DDoS attack using activity pattern of IoT devices preserving data privacy," in *Proc. 3rd Asian Conf. Innov. Technol. (ASIANCON)*, Aug. 2023, pp. 1–6.
- [49] P. S. Bouzinis et al., "StatAvg: Mitigating data heterogeneity in federated learning for intrusion detection systems," *IEEE Trans. Netw. Service Manage.*, vol. 22, no. 4, pp. 2944–2955, Aug. 2025.
- [50] N. Wang, S. Shi, Y. Chen, W. Lou, and Y. T. Hou, "FeCo: Boosting intrusion detection capability in IoT networks via contrastive learning," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 4, pp. 4215–4230, Jul. 2025.
- [51] H. Peng, C. Wu, and Y. Xiao, "FD-IDS: Federated learning with knowledge distillation for intrusion detection in non-IID IoT environments," *Sensors*, vol. 25, no. 14, p. 4309, Jul. 2025.
- [52] A. Karunamurthy, K. Vijayan, P. R. Kshirsagar, and K. T. Tan, "An optimal federated learning-based intrusion detection for IoT environment," *Sci. Rep.*, vol. 15, no. 1, p. 8696, Mar. 2025, doi: [10.1038/s41598-025-93501-8](https://doi.org/10.1038/s41598-025-93501-8).
- [53] T. S. Messerges, E. A. Dabbish, and R. H. Sloan, "Examining smart-card security under the threat of power analysis attacks," *IEEE Trans. Comput.*, vol. 51, no. 5, pp. 541–552, May 2002.
- [54] M. Wazid, A. K. Das, S. Kumari, X. Li, and F. Wu, "Design of an efficient and provably secure anonymity preserving three-factor user authentication and key agreement scheme for TMIS," *Secur. Commun. Netw.*, vol. 9, no. 13, pp. 1983–2001, Sep. 2016.
- [55] (2020). *Federated Logistic Regression*. [Online]. Available: <https://fate.readthedocs.io/en/develop/buildtemp/python/federatedml/linearmodel/logisticregression/README.html>